



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester II Examinations 2024–2025

Course Instance Codes(s)	3BA1, 4BME1, 4BMU1, 1EM1, 4BCT1, 3BMS2, 4BMS2, 3BS9, 4BS2
Exam(s)	3rd Arts, 4th Arts, 3rd Science, 4th Science, 3rd BA Education
Module Codes	MA3491/MA539
Module	Fields and Applications
Paper No	1
External Examiner	Prof. C. Roney-Dougal
Internal Examiner(s)	Dr N. Madden ★ Dr E. Sköldberg

Instructions:

Answer all FIVE questions

Duration	2 Hours
Number of Pages	3 (including this page)
Discipline	Mathematics

Requirements:

Release in Exam Venue	Yes ☒	No ☐
Release to Library	Yes ☒	No ☐
Statistical/ Log Tables:	State Examination Commission's <i>Formulae and Tables</i>	
Handout	None	
Cambridge Tables	None	
Graph Paper	None	
Log Graph Paper	None	
Other Materials	None	
Graphic material in colour	Yes ☐	No ☒

- Q1. (a) (i) State the *sphere packing bound* [2 marks]
(ii) Determine the upper bound the sphere packing bound gives on the number of code words in a binary code with block length 10 and minimum distance 5. [1 marks]
(b) (i) State the *Singleton bound*. [2 marks]
(ii) Prove the Singleton bound. [4 marks]
(iii) Determine the upper bound the Singleton bound gives on the number of codewords in a ternary code with block length 9 and minimum distance 3. [1 marks]
(c) Consider the ternary 7-ply repetition code C , i.e. the code over the alphabet $\{0, 1, 2\}$ comprising of all words of length seven containing a single symbol.
(i) Determine the parameters of C [2 marks]
(ii) Show that C is not a perfect code. [2 marks]
(iii) Show that C is an MDS code. [2 marks]
(d) Consider the field $F_4 = \{0, 1, a, b\}$ where $a^2 = b$, $b^2 = a$, and $ab = 1$. We have that the field $\mathbb{F}_2$ is a subfield of F_4 .
(i) Show that the set $\{1, a\}$ is a basis for $\mathbb{F}_4$ as a vector space over $\mathbb{F}_2$.
(ii) Suppose that V is a vector space over $\mathbb{F}_4$ with basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$. Show that the set

$$\{\mathbf{e}_1, \dots, \mathbf{e}_n, a\mathbf{e}_1, \dots, a\mathbf{e}_n\}$$

is a basis for V as a vector space over $\mathbb{F}_2$. [4 marks]

- Q2. (a) Show that the polynomial $p(x) = x^2 + x + 1 \in \mathbb{F}_5[x]$ is irreducible. [4 marks]
(b) Calculate the multiplicative inverse of $x + 3$ in $\mathbb{F}_{25}$. (Use the presentation $\mathbb{F}_{25} = \mathbb{F}_5/(p(x))$ with $p(x)$ from part (a); give the result as a polynomial of degree ≤ 1). [6 marks]
(c) Let F be a field.
(i) What is meant by a *primitive element* of F ? [2 marks]
(ii) Show that for any $a \in F$, we have that $(a^{-1})^n = (a^n)^{-1}$ [4 marks]
(iii) Show that if a is a primitive element in a field F , then so is a^{-1} . [4 marks]

- Q3. Consider the ternary code C with generator matrix

$$G = \begin{pmatrix} 0 & 2 & 2 & 1 \\ 2 & 2 & 1 & 0 \end{pmatrix}$$

- (a) Construct a generator matrix in standard form for C . [4 marks]
(b) Construct a check matrix for C . [2 marks]
(c) Determine the minimum distance of C . [2 marks]
(d) Show that C is self-dual, i.e. that $C = C^\perp$. [4 marks]
(e) (i) Construct a syndrome table for C . [4 marks]
(ii) Use the syndrome table to decode the message words 1111 and 1122. [4 marks]

- Q4. (a) What is the definition of the *Hamming code* $Ham(r, q)$? [2 marks]
- (b) Construct a parity check matrix for the Hamming code $Ham(3, 2)$. [4 marks]
- (c) Explain how a single error can be corrected in $Ham(3, 2)$. Decode the words 1111111, and 0111110. [6 marks]
- (d) State (without proof) the parameters of the Hamming code $Ham(r, q)$. [2 marks]
- (e) Show that the Hamming code $Ham(r, q)$ is a perfect code. [6 marks]
- Q5. (a) Let C be the cyclic code of length 5 over the field $\mathbb{F}_4$ with generator polynomial $x^2 + ax + 1$. (We are considering the field $\mathbb{F}_4$ with the elements $0, 1, a, b$ with $a^2 = b$, $b^2 = a$, and $ab = 1$.)
- (i) Construct a generator matrix for C . [4 marks]
- (ii) Find the check polynomial for C and hence construct a check matrix for C . [8 marks]
- (iii) Show that C is an MDS code. [4 marks]
- (b) Let C be a binary cyclic code with odd block length. Show that if C contains a word $\mathbf{u}$ of odd weight, then it contains the ‘all 1s’ word $11\dots 1$. [Hint: consider the sum of all cyclic shifts of $\mathbf{u}$.] [4 marks]